

STATS 200A Discussion 2

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Remark 2.1 (Review from last session)

$$H_n = \{n\text{'th flip is heads}\}$$

$$H_3 = \{\omega \mid \omega = _1_ \}$$

$$A = \{\text{heads occurs at least once}\} = \bigcup_{n=1}^{\infty} H_n$$

$$B = \{\text{Heads occurs before or on the 10th toss}\} = \left(\bigcup_{n=11}^{\infty} H_n \right)^c = \bigcap_{n=11}^{\infty} H_n^c$$

Exercise 2.2

$$C = \{\text{only see finitely many heads}\}$$

Solution.

$$C = \{\exists T < \infty \text{ such that you only get tails after } T\}$$

$$C = \bigcup_{T=0}^{\infty} \bigcap_{n=T}^{\infty} H_n^c$$

$$A = B \equiv A \subset B \wedge B \subset A$$

$$C = \left(\bigcap_{T=0}^{\infty} \bigcup_{n=T}^{\infty} H_n \right)^c$$

■

Exercise 2.3

$$\bigcap_{T=0}^{\infty} \bigcup_{n=T}^{\infty} H_n^c$$

Solution.

$$\bigcap_{T=0}^{\infty} \underbrace{\bigcup_{n=T}^{\infty} H_n^c}_{\text{observe tail after or on } T \text{ flip}} = D$$

Observe infinitely many tails

■

$\omega = \{01010101 \dots\}, \omega \in D, \omega \notin C$
 $D = \text{"infinitely often" (i.o.)} = \limsup_{n \rightarrow \infty} H_n^c$
 $C = \text{"all but finitely many" (abfm)} = \liminf_{n \rightarrow \infty} H_n^c$

Definition 2.4

$$\begin{aligned}
 &(\Omega, \mathcal{F}) \\
 \mathcal{F} = &\begin{cases} \emptyset \in \mathcal{F}, \Omega \in \mathcal{F} \\ A \in \mathcal{F} \Leftrightarrow A^c \in \mathcal{F} \\ A_1, A_2, A_3, \dots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F} \end{cases} \\
 &\mathbb{P} : \mathcal{F} \rightarrow [0, 1]
 \end{aligned}$$

Example 2.5

$$f(x) = x^2$$

Solution. $\mathbb{R} \rightarrow \mathbb{R}^+$ ■

$$\mathbb{P}(x) = 0, 1, 0.5, \dots, \mathbb{P}([0, 2)), \mathbb{P}([-1, 3) \cup [3, 5))$$

Definition 2.6

$$\mathbb{P} : \mathcal{F} \rightarrow [0, 1] = \begin{cases} \mathbb{P}(\emptyset) = 0 \\ A_1, A_2, A_3, \dots \in \mathcal{F} \text{ pairwise disjoint} \equiv \bigcap_i A_i = \emptyset \\ \rightarrow \sum_{n=1}^{\infty} \mathbb{P}(A_n) = \mathbb{P}(\bigcup_{n=1}^{\infty} A_n) \end{cases}$$

Example 2.7

Prove $A, B \in \mathcal{F}$ then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$

Solution.

$$\begin{aligned}
 &A, B \in \mathcal{F} \\
 \mathbb{P}(A \cup B) &= \mathbb{P}(A \cup B \setminus A) \\
 &= \mathbb{P}(A) + \mathbb{P}(B \setminus A)
 \end{aligned}$$
■

Example 2.8

$$A_1, A_2, \dots \in \mathcal{F} \rightarrow \sum_{n=1}^{\infty} \mathbb{P}(A_n) \leq \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right)$$

Solution. Let $B_1 = A_1, B_2 = A_2 \setminus A_1, \dots, B_n = A_n \setminus A_{n-1}$
 Use countable additivity since $B_1, \dots$ pairwise disjoint since $\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$
 $\mathbb{P}(A_n) = \mathbb{P}(B_n) + \mathbb{P}(A_n \setminus B_n) \rightarrow \mathbb{P}(A_n) \geq \mathbb{P}(B_n)$
 $\sum_{n=1}^{\infty} \mathbb{P}(A_n) \geq \sum_{n=1}^{\infty} \mathbb{P}(B_n)$
 $\sum_{n=1}^{\infty} \mathbb{P}(B_n) = \mathbb{P}(\bigcup_{n=1}^{\infty} B_n) = \mathbb{P}(\bigcup_{n=1}^{\infty} A_n)$ ■

Example 2.9

$$A_1 \subset A_2 \subset A_3 \subset \cdots \rightarrow \lim_{n \rightarrow \infty} \mathbb{P}(A_n) = \mathbb{P}(\bigcup_{n=1}^{\infty} A_n)$$